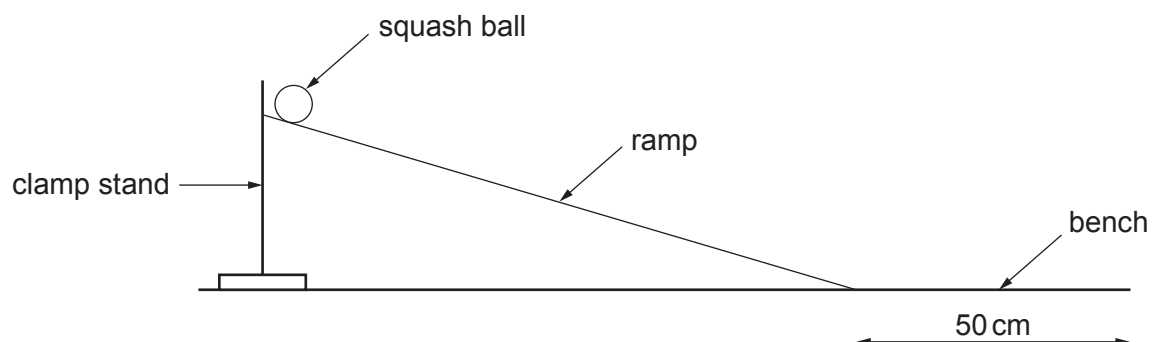


4. The diagram shows the apparatus used to find the acceleration of a squash ball as it rolls down a ramp. A stopwatch is also required.



- (a) Describe the method used in this investigation.
Include in your answer:

[6 QER]

- How you would carry out the experiment
- How to reduce the uncertainty in your results.

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- (b) The following readings were taken during the investigation.

Time to travel 50 cm along the bench = 2.0 s

Time to reach the bottom of the ramp = 2.5 s

- (i) Use the equation:

$$\text{velocity} = \frac{0.5}{\text{time to travel 50 cm along the bench}}$$

to calculate the velocity of the ball travelling along the bench.

[2]

velocity = m/s

- (ii) Use the equation:

$$\text{acceleration} = \frac{\text{velocity along the bench}}{\text{time to reach the bottom of the ramp}}$$

to calculate the acceleration of the ball down the ramp.

[2]

acceleration = m/s²

